

Convergence of Electronics with Fabrics

e-textiles are an important part of wearable computing. Also called 'e-fabrics,' these have various electronic components and interconnections woven into them for the desired functionality. Here is an overview of their design, usage and areas of research

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Electronic gadgets are practically becoming slim. Even the clothes you wear can become your intimate companions, enhancing your functional ability. There are 'e-clothes' for almost every requirement that can make life easier, funnier, smarter and even healthier.

e-textiles, also called 'e-fabrics,' have various electronic components and interconnections woven into them. These fabrics can then be mounted on readymade garments like shirts and jackets to perform specific functions. One important key to success of any electronic product is its adaptability with existing products. That is what e-textiles aim at.

These fabrics are highly flexible. The electronic components woven into them have smaller and thinner sizes that cannot be achieved with existing electronic manufacturing techniques. Also, the components and interconnections are less visible. They do not trouble the wearer and are less affected by environmental factors like moisture. The metallic fibres and textile fibres together form conductive fibres.

e-clothes can be worn in everyday situations where currently available computer applications would trouble the user. e-textiles can more easily adapt to changes in the computational and sensing requirements of an application. These also provide better power

management and context awareness. In fact, in the future you may find e-fibres on walls, carpets and wall hangings of homes to increase their capabilities.

Design methodology

Typical challenges before e-textile designers are how to make connections between woven wires and how to mount electronic components in these fabrics. For several e-textile projects,

pins should be wider. Usually, an expression diagram is used in the circuit design of e-clothes.

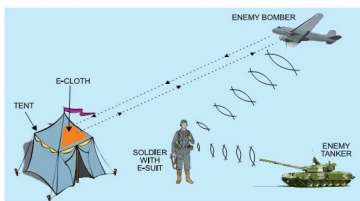
One of the key requirements of e-textile components is interactive properties. That is, they should be able to change structure, colour and temperature in different ways in response to environmental stimuli.

Use of sensors in the circuit depends upon the application for which e-clothes are meant. For example, in automotive and road traffic applications, several factors play an important role. These include acceleration, temperature, humidity and light. Sensors capable of sensing such parameters should be selected.

In case of data transfer requirement, components having Bluetooth communication interface should be used. For wireless data communication, components having wireless data transfer capability should be used.

Focus areas

e-textiles are a must in situations where wearable computing is required. For instance, e-clothes are a perfect fit for soldiers. These are designed to handle bad weather and tough terrains, and can be worn comfortably because of their light weight. They can help maintain proper body temperature in freezing cold. Allowing flexible computing, e-clothes help soldiers detect and locate enemy vehicle movements. e-fabrics can be potentially fabricated



Use of e-textiles in military

the LilyPad motherboard is used. It offers a wide range of built-in functions.

Some of the key requirements for e-fabrics are that these should be able to withstand the applied stresses and be easily washable. They should be low in cost, easy to manufacture and unfading. Also, e-textiles should be fault-tolerant and self-configurable.

One must know how many processors and sensors are required for a particular application and how to put them between the fibres. Most of the microchips today are packaged with hundreds of pins. But for e-textiles microchips require only four to five pins. Further, space between these